Preface

Microservices are an approach to distributed systems that promote the use of finely grained services that can be changed, deployed, and released independently. For organizations that are moving toward more loosely coupled systems, with autonomous teams delivering user-facing functionality, microservices work incredibly well. Beyond this, microservices provide us with a huge number of options for building out systems, giving us a lot of flexibility to ensure our system can change to meet the needs of the users.

Microservices aren't without significant downsides, though. As a distributed system, they bring a host of complexity, much of which may be new even to experienced developers.

The experiences of people all over the world, along with the emergence of new technologies, are having a profound effect on how microservices are used. This book brings these ideas together, along with concrete, real-world examples, to help you understand whether microservices are right for you.

Who Should Read This Book

The scope of *Building Microservices* is broad, as the implications of microservice architectures are also broad. As such, this book should appeal to people interested in aspects of design, development, deployment, testing, and maintenance of systems. Those of you who have already embarked on the journey toward finer-grained architectures, whether for a greenfield application or as part of decomposing an existing, more monolithic system, will find plenty of practical advice to help you. This book will also help those of you who want to know what all the fuss is about, so that you can determine whether microservices are right for you.

Why I Wrote This Book

At one level, I wrote this book because I wanted to make sure that the information in the first edition remained up to date, accurate, and useful. I wrote the first edition because there were really interesting ideas that I wanted to share. I was fortunate that I was in a

place where I had the time and support to write the first edition, and that I could do so from a fairly unbiased viewpoint because I didn't work for a big technology vendor. I wasn't selling a solution, and I hoped I wasn't *selling* microservices either—I just found the ideas fascinating, and I loved unpacking the concept of microservices and finding ways of sharing it more broadly.

When I really break it down, I wrote a second edition for two main reasons. Firstly, I felt I could do a better job this time. I've learned more, and I am hopefully a bit better as a writer. But I also wrote this second edition because I played a small part in helping these ideas hit the mainstream, and as such I have a duty of care to make sure they are presented in a sensible, balanced way. Microservices have become, for many, the default architectural choice. This is something that I think is hard to justify, and I wanted a chance to share why.

This book isn't *pro* microservices, nor is it *anti* microservices. I just want to make sure that I've properly explored the context in which these ideas work well and shared the problems they can cause.

What's Changed Since the First Edition?

I wrote the first edition of *Building Microservices* in around a year, starting at the beginning of 2014, with the book being released in February 2015. This was early in the microservices story, at least in terms of the wider industry's awareness of the term. Since then, microservices have gone mainstream in a way that I couldn't have predicted. With that growth has come a much wider set of experiences to draw on, and more technology to explore.

As I worked with more teams in the wake of the first edition, I started to refine my thinking around some of the ideas associated with microservices. In some cases, this meant that ideas that were only at the periphery of my thinking, like information hiding, started to become more clear as foundational concepts that needed to be better highlighted. In other areas, new technology presented both new solutions and new problems for our systems. Seeing so many people flock to Kubernetes in the hope that it would solve all their issues with microservice architectures certainly gave me pause for thought.

Additionally, I'd written the first edition of *Building Microservices* to provide not just an explanation of microservices, but also a broad overview of how this architectural approach changes aspects of software development. So as I looked more deeply at aspects

around security and resiliency, I found myself wanting to go back and expand on those topics that are increasingly important to modern software development.

Thus in this second edition, I spend more time sharing explicit examples to better explain the ideas. Every chapter has been reexamined, and every sentence reviewed. Not a lot of the first edition remains in terms of concrete prose, but the ideas are all still here. I've tried to be clearer in my own opinions, while still recognizing that there are often multiple ways of solving a problem. This has meant expanding the discussion of inter-process communication, which is now split across three chapters. I also spend more time looking at the implications of technologies such as containers, Kubernetes, and serverless; as a result, there are now separate build and deployment chapters.

My hope had been to create a book around the same size as the first edition, while finding a way to pack in more ideas. As you can see, I've failed to achieve my goal—this edition is bigger! But I think I've succeeded in being clearer in how I articulate the ideas.

Navigating This Book

This book is primarily organized in a topic-based format. I have structured and written the book as though you will read it from beginning to end, but of course you may want to jump to the specific topics that most interest you. If you do decide to dive right into a specific chapter, you may find the Glossary toward the end of the book helpful in explaining new or unfamiliar terms. On the topic of terminology, I use microservice and service interchangeably throughout the book; you can assume that the two terms refer to the same thing unless I explicitly state otherwise. I also summarize some of the book's key advice in the Bibliography if you *really* just want to jump to the end—just remember you'll be missing a *lot* of detail if you do that!

The main body of the book is broken into three separate parts—Foundation, Implementation, and People. Let's look at what each part covers.

Part I, "Foundation"

In this part, I detail some of the core ideas behind microservices.

Chapter 1, "What Are Microservices?"

This is a general introduction to microservices in which I briefly discuss a number of topics that I'll expand on later in the book.

Chapter 2, "How to Model Microservices"

This chapter examines the importance of concepts like information hiding, coupling, cohesion, and domain-driven design in helping find the right boundaries for your microservices.

Chapter 3, "Splitting the Monolith"

This chapter provides some guidance on how to take an existing monolithic application and break it apart into microservices.

Chapter 4, "Microservice Communication Styles"

The last chapter in this part offers a discussion of the different types of microservice communication, including asynchronous versus synchronous calls and request-response and event-driven collaboration styles.

Part II, "Implementation"

Moving from higher-level concepts to implementation detail, in this part we look at techniques and technologies that can help you get the most out of microservices.

Chapter 5, "Implementing Microservice Communication"

In this chapter we take a deeper look at the specific technologies used to implement inter-microservice communication.

Chapter 6, "Workflow"

This chapter offers a comparison of sagas and distributed transactions and discusses their usefulness in modeling business processes involving multiple microservices.

Chapter 7, "Build"

This chapter explains mapping from a microservice to repositories and builds.

Chapter 8, "Deployment"

In this chapter you'll find discussion of the myriad options around deploying a microservice, including a look at containers, Kubernetes, and FaaS.

Chapter 9, "Testing"

Here I discuss the challenges of testing microservices, including the issues caused by end-to-end tests, and how consumer-driven contracts and in-production testing can help.

Chapter 10, "From Monitoring to Observability"

This chapter covers the shift from focusing on static monitoring activities to thinking more broadly about how to improve the observability of microservice architectures, along with some specific recommendations regarding tooling.

Chapter 11, "Security"

Microservice architectures can increase the surface area of attack but also give us more opportunity to defend in depth. In this chapter, we explore this balance.

Chapter 12, "Resiliency"

This chapter offers a wider look at what resiliency is and at the part that microservices can play in improving the resiliency of your applications.

Chapter 13, "Scaling"

In this chapter I outline the four axes of scaling and show how they can be used in combination to scale a microservice architecture.

Part III, "People"

Ideas and technology don't mean anything without the people and organization to back them up.

Chapter 14, "User Interfaces"

From moving away from dedicated frontend teams to the use of BFFs and GraphQL, this chapter explores how microservices and user interfaces can work together.

Chapter 15, "Organizational Structures"

This penultimate chapter focuses on how stream-aligned and enabling teams can work in the context of microservice architectures.

Chapter 16, "The Evolutionary Architect"

Microservice architectures aren't static, so your view of system architecture may need to change—a topic that this chapter explores in depth.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

`Constant width`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

This element indicates a warning or caution.

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Much of this book was written during a global pandemic, which as I write these acknowledgments is still ongoing. It may not mean much, but I want to extend my thanks to the NHS here in the UK, and to all the people worldwide who are keeping us safe by working on vaccines, treating the sick, delivering our food, and helping in a thousand other ways I wouldn't understand. This is also for all of you.

This second edition wouldn't exist without the first, so I'd like to reiterate my thanks to everyone who helped me during the challenging process of writing my first book, include the technical reviewers Ben Christensen, Martin Fowler, and Venkat Subramaniam; James Lewis for our many enlightening conversations; the O'Reilly team of Brian MacDonald, Rachel Monaghan, Kristen Brown, and Betsy Waliszewski; and excellent review feedback from readers Anand Krishnaswamy, Kent McNeil, Charles Haynes, Chris Ford, Aidy Lewis, Will Thames, Jon Eaves, Rolf Russell, Badrinath Janakiraman, Daniel Bryant, Ian Robinson, Jim Webber, Stewart Gleadow, Evan Bottcher, Eric Sword, and Olivia Leonard. And thank to Mike Loukides, I guess, for getting me into this mess in the first place!

For the second edition, Martin Fowler once again returned as technical reviewer and was joined by Daniel Bryant and Sarah Wells, who were generous with their time and feedback. I'd also like to thank Nicky Wrightson and Alexander von Zitzerwitz for helping push the technical review over the finishing line. On the O'Reilly front, the whole process was overseen by my awesome editor, Nicole Taché, without whom I would certainly have gone insane; Melissa Duffield, who seems to do a better job of managing my workload than I do; Deb Baker, Arthur Johnson, and the rest of the production team (I'm sorry I don't know all your names, but thank you!); and Mary Treseler, for steering the ship through some difficult times.

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I wrote the bulk of this book throughout the whole of 2020 and in the first half of 2021. I used Visual Studio Code on macOS for most of the writing, although on rare occasions I also used Working Copy on iOS. OmniGraffle was used to create all the diagrams in the book. AsciiDoc was used to format the book and on the whole was excellent, with O'Reilly's Atlas toolchain doing the magic of making a book appear on the other side.

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